
3.4.4 In respiration, glycolysis takes place in the cytoplasm and the remaining steps in the mitochondria. ATP synthesis is associated with the electron transfer chain in the membranes of mitochondria.

Aerobic respiration	Aerobic respiration in such detail as to show that • glycolysis takes place in the cytoplasm and involves the oxidation of glucose to pyruvate with a net gain of ATP and reduced NAD • pyruvate combines with coenzyme A in the link reaction to produce acetylcoenzyme A • in a series of oxidation-reduction reactions the Krebs cycle generates reduced coenzymes and ATP by substrate-level phosphorylation, and carbon dioxide is lost • acetylcoenzyme A is effectively a two carbon molecule that combines with a four carbon molecule to produce a six carbon molecule which enters the Krebs cycle • synthesis of ATP by oxidative phosphorylation is associated with the transfer of electrons down the electron transport chain and passage of protons across mitochondrial membranes.
Anaerobic respiration	Glycolysis followed by the production of ethanol or lactate and the regeneration of NAD in anaerobic respiration.
